Introduction to Linear Modeling

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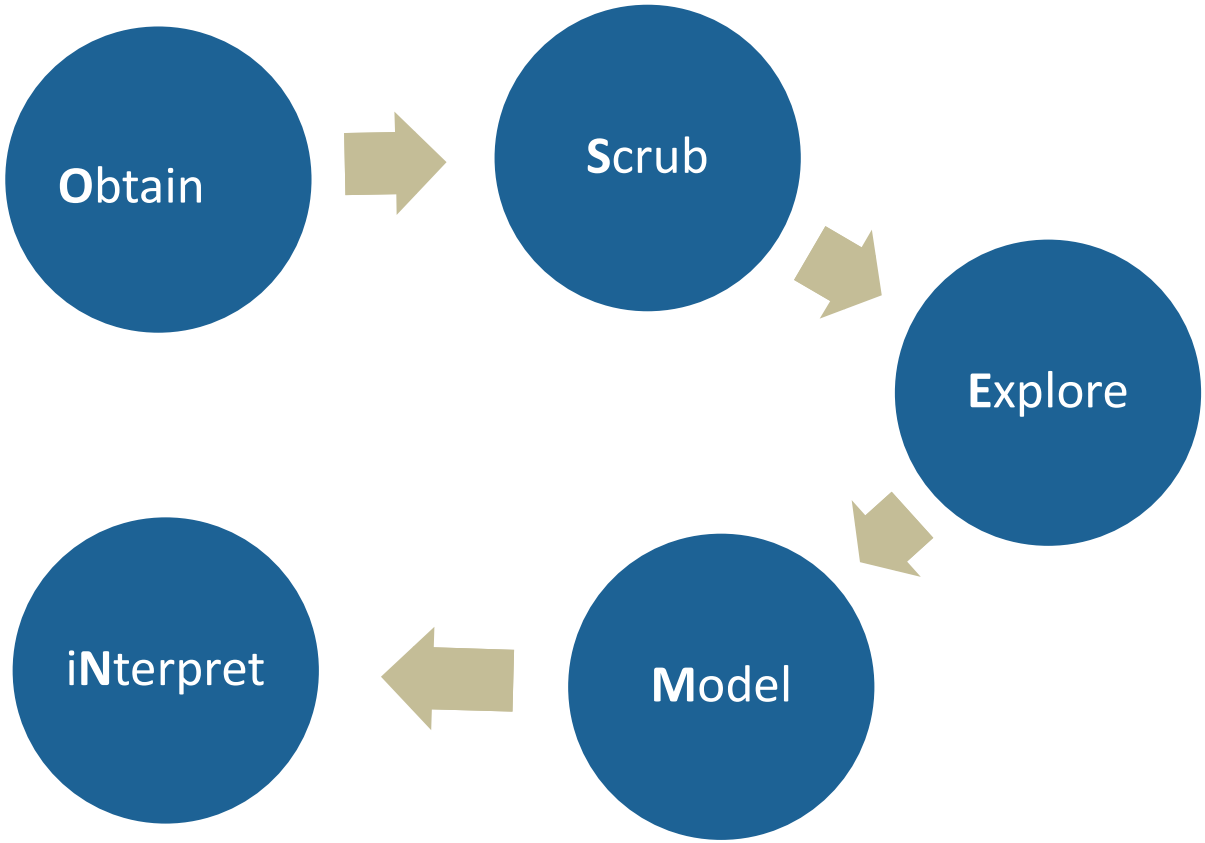
Machine Learning Overview

The ML/Data Science Process
OSEMN (Rhymes With Possum)



Hilary Mason and Chris Wiggins, 2010 - http://www.dataists.com/2010/09/a-taxonomy-of-data-science

Trad Statistical Analysis vs. Machine Learning / Data Science

Statistical Analysis

- Often focused on confirming a hypothesis in light of some theory
- Data arrives with provenance, screening & cleaning are trivial
- Measurement strategy planned in advance, thoughtful choice of constructs and scales
- Metric variables often follow the normal distribution

Data Mining / Data Science

- More exploratory, looking for novel patterns and needles in haystacks
- Data may arrive with little provenance, may require extensive diagnosis and restructuring
- No planned measurement strategy: work with what you have available
- Mixture of textual, categorical, & metric, no presumption of distribution

Machine Learning

- Enables computers to learn new things without explicit programming
- Part of artificial intelligence, i.e. creating intelligent machines

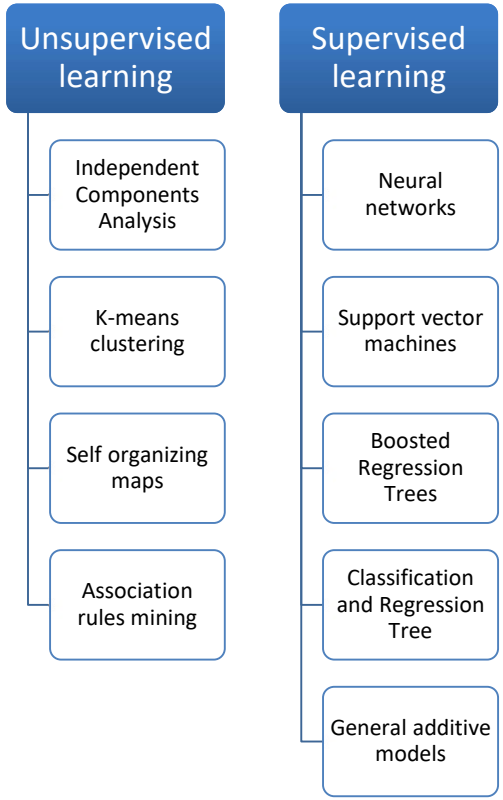
- **Supervised**: learning relationships between explanatory vars ("features") and a pre-labeled outcome from a "training" dataset to apply to a testing set.

- **Unsupervised**: finding patterns in data with no pre-existing "labels" ~ clustering or association rules

- **Reinforcement**: game-like trial-and-error actions to maximize "reward"

Machine Learning Techniques

Unsupervised learning includes a variety of machine learning techniques that do not use a dependent variable, but rather look for patterns solely among "independent" variables.



Supervised learning is parallel in concept to the predictive statistical techniques used by many social science researchers, such as linear regression, but without the restriction of only exploring linear relationships.

Another form of learning is known as "**reinforcement learning**." Evolution of these models depends on success/failure cues from real or simulated environments.

Supervised, unsupervised, or reinforcement learning?

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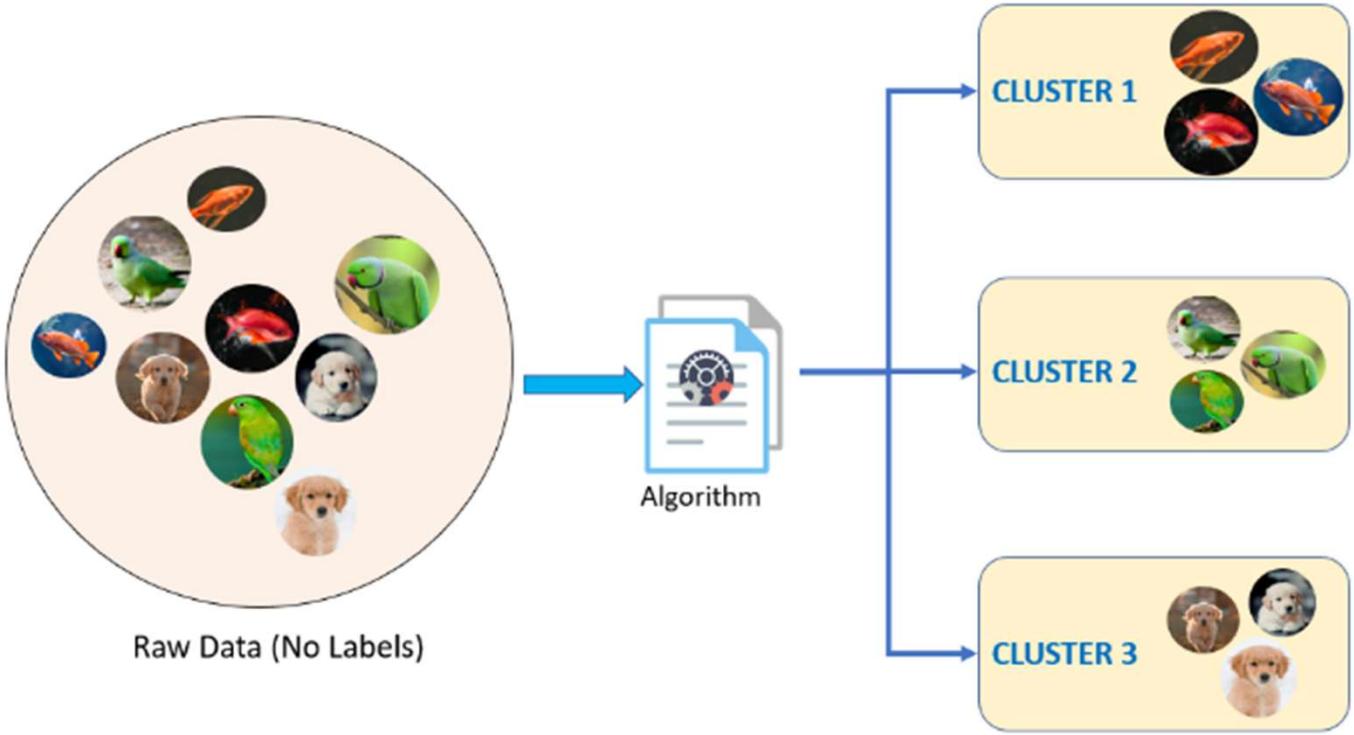
Supervised, unsupervised, or reinforcement learning?



Supervised ML

Two types of prediction problems

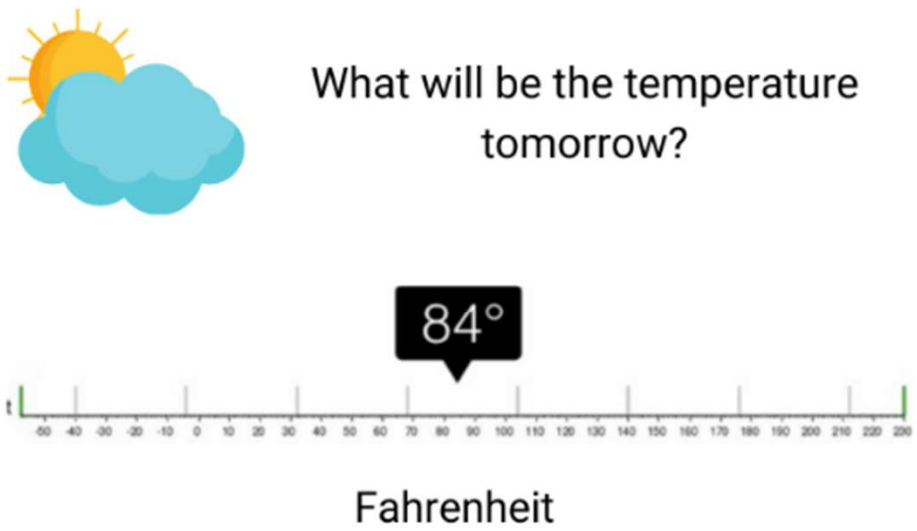
Classification problems

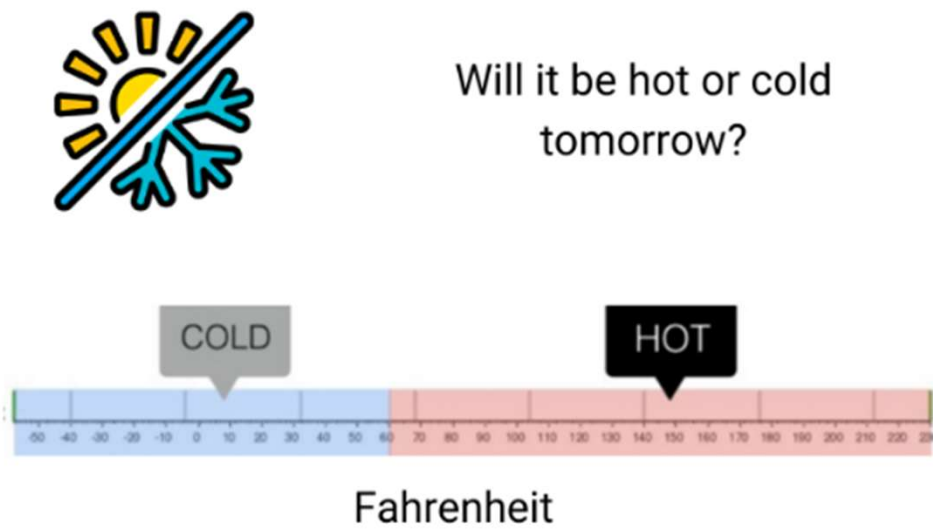
- **Predicting membership in two or more categories** based on a set of predictors (model features)
  - **Examples of criteria to predict**:
    Medical diagnosis,
    Employment outcomes (e.g., hiring),
    Financial outcomes (e.g., loan default)

Regression problems

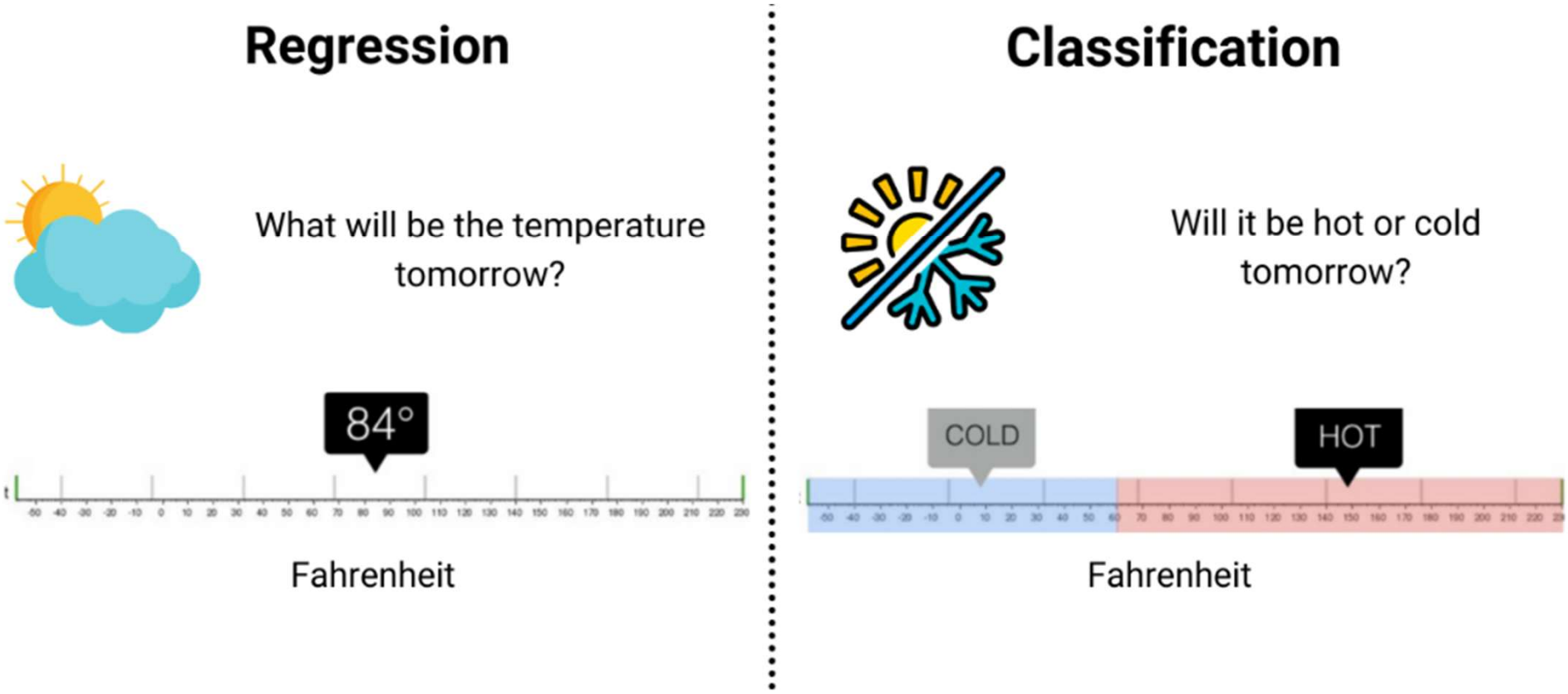
- **Predicting a continuous numeric outcome** based on on a set of predictors
  - **Examples of criteria to predict:**
    sales volume,
    employee engagement,
    customer satisfaction

Classification vs. Regression





Classification vs. Regression



Linear Modeling Overview

Linear Modeling

“Does X influence Y?” .

Arriving at a Good Prediction

- Given the following data, what is your best guess about test score #10?

Exam #	Exam Score
1	30
2	90
3	45
4	50
5	30
6	37
7	60
8	68
9	42
10	**???**

Arriving at a Good Prediction

- We only have one variable to work with – the variable we are trying to predict:

Just a descriptor

Exam #	Exam Score
1	30
2	90
3	45
4	50
5	30
6	37
7	60
8	68
9	42
10	???

Actual variable

Introducing a Predictor Variable

- Perhaps the time a student spends studying can help explain some part of their test score.
- Different names, same meaning:

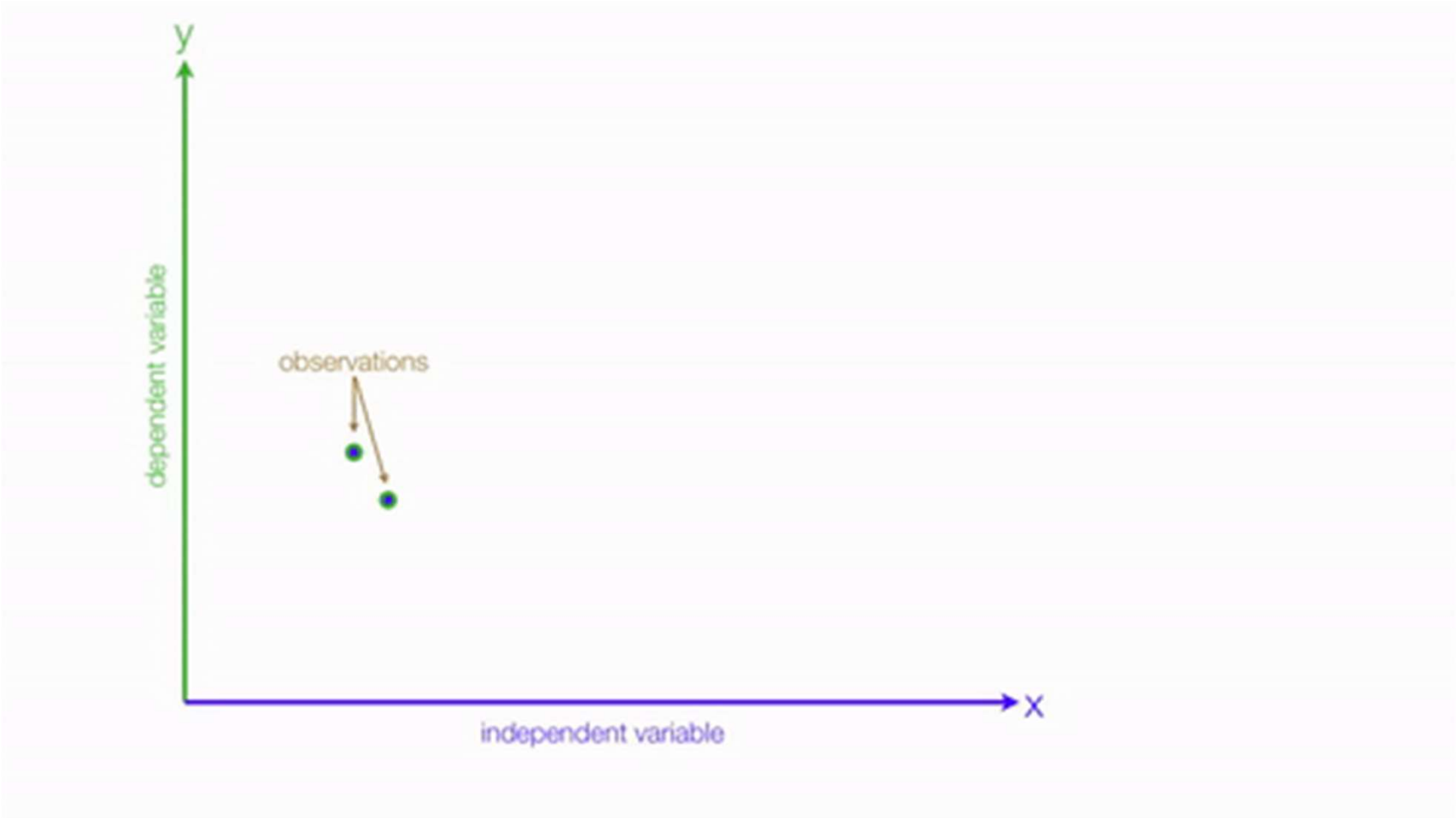
X	Y
Predictor variable	Predicted variable
Independent variable (IV)	Dependent variable (DV)
Explanatory variable	Response variable
Regressor	Regressand
Input variable	Outcome variable

Introducing a Predictor Variable

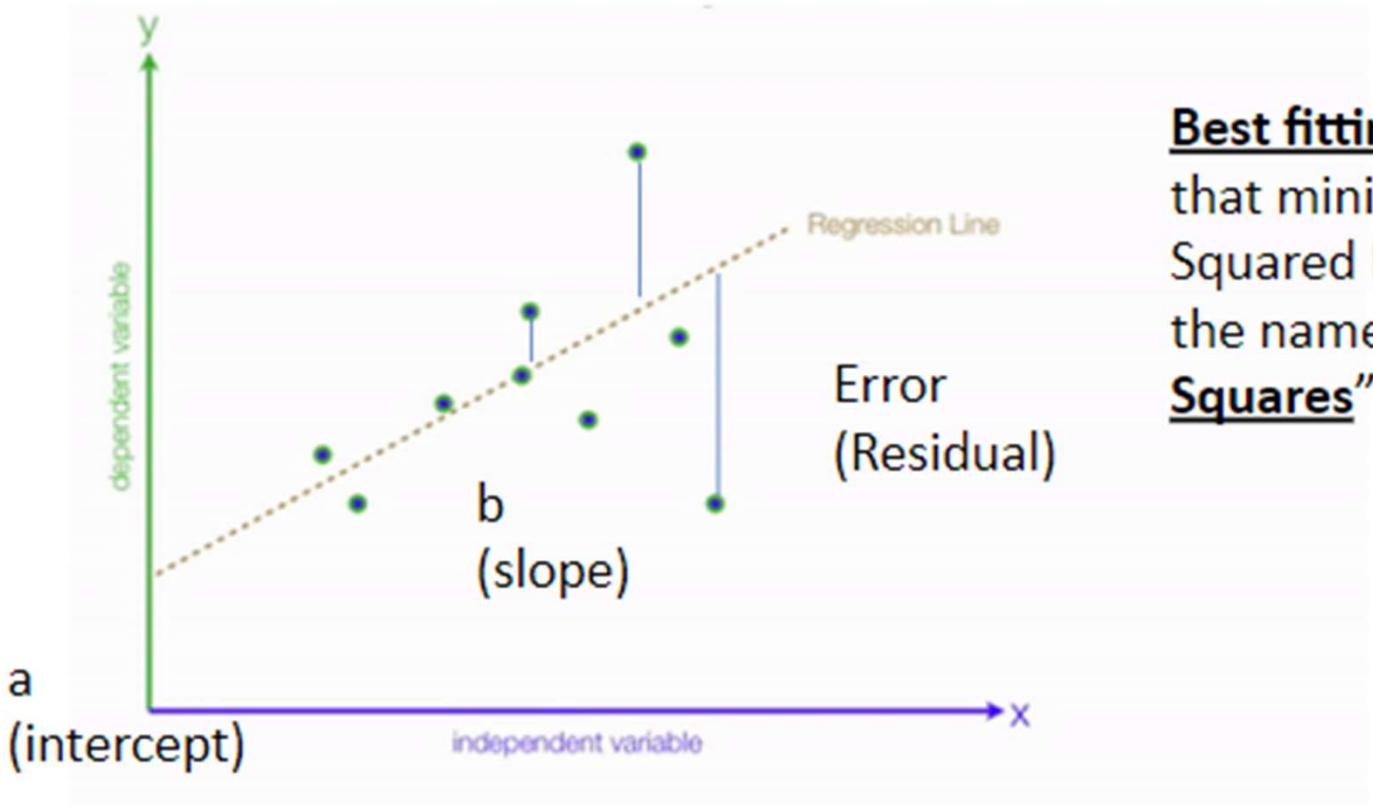
- Perhaps the time a student spends studying can help explain their test score:

Exam #	Exam Score	Time Studying (min)
1	30	10
2	90	100
3	45	60
4	50	80
5	30	12
6	37	8
7	60	110
8	68	90
9	42	30
10	**???**	**45**

Plotting the relationship

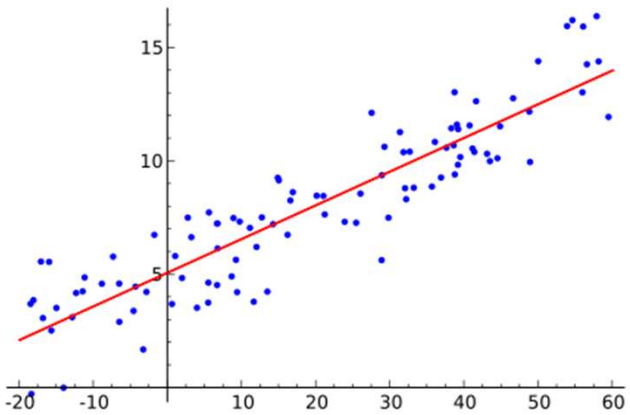


Prediction errors

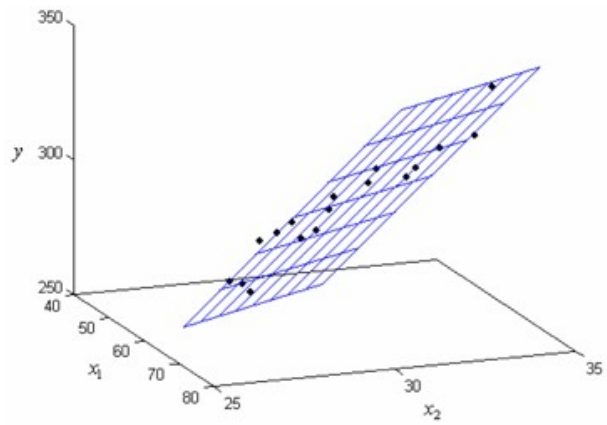


Best fitting line = line that minimizes the Squared Errors, hence the name "**Least Squares**"

Linear Relationships in 2+ Dimensions



Two dimensions: easy



Three dimensions: still good

?

Four dimensions?

Image credit: Sewaqu - Public Domain, https://commons.wikimedia.org/w/index.php?curid=11967659

A Model to Predict Exam Scores

Predicting a student's exam score using three pieces of information?

Data collection:

- Measure three predictors, using tests or surveys with multiple students (e.g., n=120)

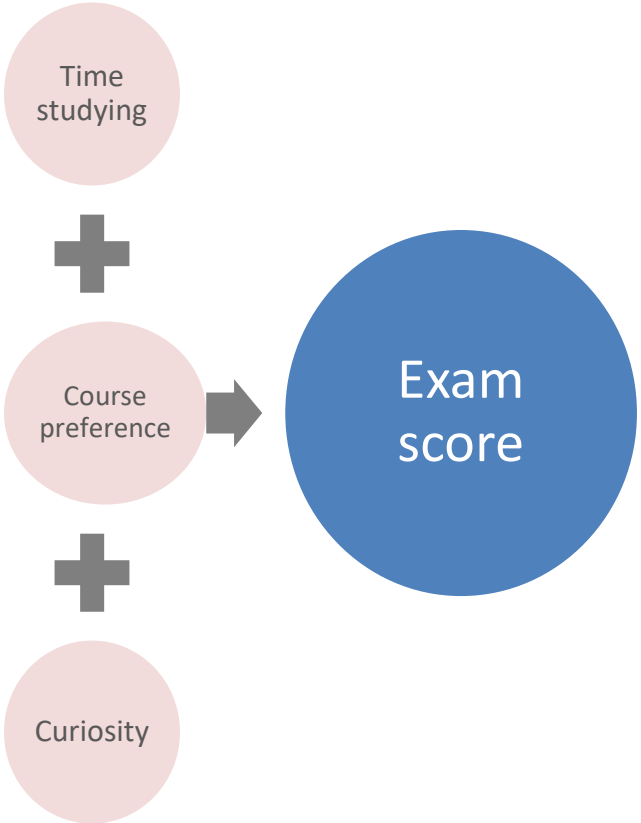
- Then at the end of the semester we would have four pieces of information:
  - The outcome (Exam score)
  - The three predictors (time studying, course preference, & curiosity).

Model creation:

- Using linear regression, calculate coefficients for each predictor to make an equation:

ExamScore = Intercept + (B1 * TimeStudying) + (B2 * CoursePreference) + (B3 * Curiosity)

Y = a + bX



Regression Terminology

- Outcome/**dependent variable**: what we are trying to predict
- Predictor/**independent variable**: one of the variables we use to predict the criterion; there are usually multiple predictors
- Coefficients/**weights**: the strength of prediction for each predictor; sometimes also called B-weights (or the standardized version is called a beta-weight)
- Regression equation: the result of the regression analysis in the form of an algebraic equation
  - $Y = MX + B$

    or...
  - $\hat{Y} = B_1X_1 + B_2X_2 + B_3X_3 + \ \ldots$
  - Y-hat is the predicted Y, subscripts on Bs and Xs refer to predictor number

Coding demo:
tinyurl.com/ist8demo